

OMAR ELWEKEIL

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Education

Egypt-Japan University for Science and Technology
Bachelor of Science in Computer Science and Information Technology

Sep 2023 – July 2027
Alexandria, Egypt

Experience

Data analyst intern (DEPI)

Nov 2024 – May 2025

- Utilized Python and SQL to clean, analyze, and interpret datasets for actionable business insights.
- Developed interactive dashboards in Tableau and Power BI to visualize key performance metrics.
- Automated data processing tasks in Excel, improving reporting efficiency and accuracy.

Technologies: Python, SQL, Tableau, Power BI, Excel

Machine Learning (Coursera)

Jul 2025 – Aug 2025

- Gained solid foundations in supervised, unsupervised, and deep learning techniques.
- Implemented regression, classification, clustering, and ensemble (Boosting) models.
- Built and trained neural, convolutional, and generative models using TensorFlow and Keras.
- Strengthened understanding of model tuning, regularization, and evaluation.

Technologies: scikit-learn, TensorFlow/Keras

Computer Vision intern (NTI)

Aug 2025 – Sep 2025

- Applied Convolutional Neural Networks (CNNs) for image processing and feature extraction.
- Implemented transfer learning using modern architectures such as ResNet and DenseNet.
- Utilized YOLO models for object detection, segmentation, and tracking.
- Explored face recognition and verification systems.
- Built multiple practical projects applying computer vision and deep learning techniques.

Technologies: TensorFlow/Keras, PyTorch, OpenCV, YOLO

Problem Solving

Currently

869 Rated on CodeForces

Projects

Superstore Sales Performance Analysis (Python, Tableau)

Apr 2025

- Cleaned and transformed raw sales data, engineering new features such as predicted demand and profitability metrics.
- Applied predictive modeling and time-series analysis to forecast trends and identify top-performing product categories.
- Designed interactive Tableau dashboards with dynamic filters and KPIs to enable data-driven decision-making.

2D & 3D Convolution Optimization (C++)

May 2025

- Implemented naive 2D/3D convolution algorithms in C/C++ and profiled baseline execution times.
- Applied hardware-aware optimizations: SIMD vectorization, loop unrolling, cache blocking, and OpenMP multithreading.
- Achieved 4.8× speedup for 3D convolution and 3.2× for 2D through cache-efficient memory access patterns.

Priority-Based CPU Process Scheduler (C++)

May 2025

- Designed a scheduler for dynamic process generation (priority/arrival time) using hybrid priority + Round Robin logic.
- Developed Gantt chart visualization to display CPU allocation timelines and preemption events.
- Quantified efficiency via waiting time, turnaround time, and throughput metrics.

AI Gym Pose Estimation App (Flutter, Python, YOLOv8) (Team project)

Sep 2025

- Built a Flutter-based application that uses a live camera feed to evaluate user exercise form and provide instant feedback on correctness.
- Integrated a pre-trained YOLOv8 pose model for keypoint detection, movement tracking, and exercise classification.
- Optimized model inference and UI responsiveness to ensure smooth real-time feedback during workouts.

Technical Skills

Programming: Python, C++, JavaScript, HTML/CSS

AI & Machine Learning: Data Preprocessing & Feature Engineering, scikit-learn, TensorFlow/Keras, PyTorch, OpenCV, YOLO

Data Science & Analysis: SQL, Pandas, NumPy, Tableau, Power BI, Excel

Development & Deployment: Streamlit, Git/GitHub

Tools & Environments: VS Code, PyCharm, Google Colab

Soft Skills

Team Leadership

Self Learning